

Hacettepe University | Computer Engineering CMP 720 EMBEDDED SYSTEMS DESIGN

Spring 2026

Midterm Project Peer Evaluation Form

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Instructions

For each project, briefly summarize the work and provide your evaluation **individually (not as a group)**. Your response should reflect your understanding of the problem, methodology, and evaluation plan, and include critical insights, strengths, limitations, and possible improvements. Skip your own group's evaluation and simply state that it is your group.

Students are required to evaluate only the projects presented within their own class group (Hacettepe or ASELSAN).

Guiding points (address in one coherent paragraph):

- What problem is being solved, why is it important, and what are the most important embedded system design considerations for this problem (e.g., timing, reliability, energy, safety, security, scalability, etc.)?
- What is the main idea of the proposed methodology?
- What do you think are the strengths and limitations of the approach?
- Is there anything you would improve or do differently?
- What is the most critical or challenging component?
- Is the evaluation plan sufficient? What, if anything, is missing?

Group 1

Umut Kalman, Eren Karadağ

Project Title: *Context-Aware Deadline Scheduling Under Memory Contention for Embedded Real-Time Systems*

Group 1's project addresses a real and somewhat subtle problem, standard real-time schedulers like EDF are theoretically proven to work, but in practice they ignore what happens at the memory level. When multiple tasks compete for the same memory bus, they interfere with each other in ways the scheduler never accounted for, causing deadline misses that shouldn't happen on paper. The goal is to fix this without touching the kernel or requiring special hardware just a user-space wrapper on top of FreeRTOS. The core idea is: before running a high-memory task, the wrapper checks whether there is enough slack time to delay it slightly, so it doesn't collide with another memory-heavy task running back to back. If there is slack, it reorders; if not, it falls back to plain EDF. The penalty model they use to estimate contention is simple just multiplying two memory intensity values which is lightweight enough for a constrained microcontroller, though I wasn't sure how accurately it reflects real cache behavior. What I found interesting is the validation pipeline. They first profile worst-case execution times on real hardware using DWT cycle counters, then use an offline Ant Colony Optimization solver to find the theoretical best schedule, and finally compare their online wrapper against both that baseline and pure EDF. That three-way comparison is a solid evaluation approach. A few things I was uncertain about after the presentation. The penalty model assumes memory contention only happens between consecutive tasks upon preemption, which seems like a simplification, I'd be curious how it holds up under more complex workloads. Also, the system is only tested with three tasks, and whether the approach scales to more realistic task sets wasn't entirely clear. Overall though, the problem is well-motivated and the solution feels pragmatic and deployment-friendly.

Group 2

Yusuf Çağrı Ögüt

Project Title: *Hybrid Cellular-LoRa Network Based Agriculture Monitoring and Control Application*

Group 2's project is about remotely monitoring and controlling an off-grid hobby garden the kind of situation where someone has a vegetable garden far from their home, with no WiFi available, and needs to know if the soil is dry or a valve has failed without physically going there. The solution is a two-tier system: a cellular (SMS) layer for wide-area communication between the user's phone and a central controller, and a local LoRa radio network connecting that controller to sensor/actuator nodes buried in the garden. No internet required at any point. I found interesting that power engineering. The target is around 1 mAh per day per node, which would give roughly 100 days of battery life on a standard 3000 mAh pack. To hit this, the nodes spend 99 percent of their time in deep sleep, waking up for only 5 seconds every 20 minutes to sense and transmit. The other interesting thing is the pipe choice, they use latched valves, which draw a large current for just 50 milliseconds to switch state, and then require zero power to hold that state open or closed. This means the irrigation valves contribute almost nothing to the daily energy budget, which is a smart hardware-level decision rather than a software trick. The security consideration also interesting, the central controller enforces a caller ID whitelist, so only pre-approved phone numbers can send commands. It's simple but effective for this use case. There's also a safe state mechanism where valves automatically close if LoRa synchronization is lost for more than an hour, preventing both overwatering and leaving valves stuck open unattended. The related work comparison against Enock et al. and Heble et al. makes the positioning clear, both prior works depend on internet connectivity or cloud infrastructure, which simply doesn't exist in the target environment. The gap this project fills is genuine. A few things I'd be curious about after the presentation. The 95 percent packet delivery ratio target is measured at 50 meters with buried nodes and elevated antennas, but the soil moisture level significantly affects LoRa signal attenuation, and wet soil is the worst case. How much PDR degrades right after irrigation when the soil is saturated wasn't addressed. Also, SMS latency is controlled by the cellular network, not the device so there could be situations where the network delays a command beyond the 60-second target through no fault of the hardware.

Group 3

Yusuf Tahir Orhan

Project Title: *Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm*

Group 3's project is about a routing algorithm called DeFT, designed for a relatively new type of chip architecture called 2.5D integration. The basic idea of 2.5D is stacking multiple smaller chiplets on an interposer instead of making one giant chip — it's more modular and cost-effective. The problem is that routing data between these chiplets is tricky: packets have to go "up" from one chiplet, travel across, then go "down" into another, and this pattern can create deadlocks where packets all end up waiting for each other indefinitely. On top of that, the vertical links connecting chiplets are physically fragile and can fail permanently due to heat or manufacturing defects. DeFT addresses both problems at once deadlock prevention and fault tolerance. The first is a set of three strict rules about which virtual networks packets are allowed to use and which routing directions are forbidden, which mathematically prevents deadlock cycles from forming. The second is a set of lookup tables stored in the boundary routers that dynamically select alternative vertical links when some have failed. The whole thing is implemented in a cycle-accurate simulator called Noxim, which simulates how the hardware would actually behave clock by clock. What I found interesting is the framing. This is explicitly described as a replication study, meaning the goal isn't to invent something new but to faithfully reproduce the original DeFT paper's results and use that as a baseline. That's sounds like an honest and methodologically sound approach. A few things were less clear to me. The lookup table approach for fault routing sounds straightforward, but I'd be curious how large these tables get as you add more chips. The presentation briefly mentions scalability as a concern but doesn't go into detail. Also, since results haven't been presented yet and the evaluation is still expected, it's hard to judge how well the implementation actually matches the original paper's numbers. That's the most critical missing piece right now.

Group 4

Mehmet Yusuf Sezgi, Rana Elif Albayrak

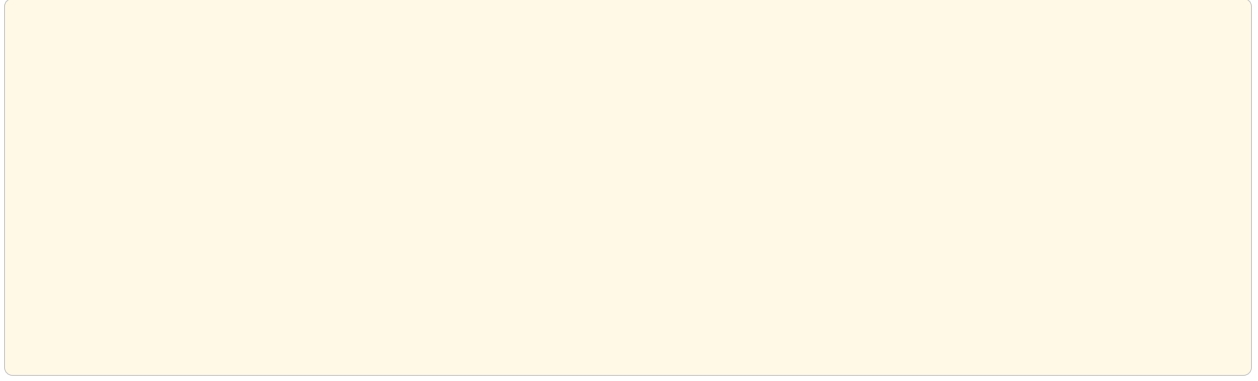
Project Title: *Real-Time CAN Decoder*

Group 4's project is about reading and making sense of CAN bus data from a vehicle in real time, using a small embedded device. CAN is basically the internal communication network that cars use and all the electronic control units like the engine, brakes, and sensors constantly talk to each other over this bus, but the raw messages are just binary frames with no obvious meaning. The goal here is to receive those frames, decode them into something meaningful like RPM or speed, and log them to an SD card, all on a resource-constrained microcontroller. What I found interesting about this project is how group 4 describe concrete the engineering constraints are. Frames arrive every 250-300 microseconds, the MCU has only 128 KB of SRAM, and the CAN controller and the SD card share the same SPI bus meaning if you're writing to the SD card and a CAN frame arrives at the same time, something has to give. Their solution is a two-stage buffering approach: a fast raw buffer absorbs incoming frames without blocking, and a second decoded buffer handles the slower SD write operations in batches. This is a sensible design that directly addresses the timing asymmetry between fast input and slow output. The software runs on RTOS with task priorities assigned by urgency, acquisition gets highest priority, SD logging gets lowest — which makes sense for me for keeping the critical path clean. One honest limitation they acknowledged is that they don't have access to an official DBC file, which is essentially the dictionary that maps CAN IDs to signal meanings. Without it, they had to reverse-engineer signal interpretations experimentally by driving the car and correlating CAN traffic with observed behavior. This is doable but inherently uncertain — there could be IDs they misidentified or missed entirely, and it limits how generalizable the decoder is to other vehicles. In previous approaches like CANMatch or CANClassify either require offline analysis or machine learning with high compute, making them unsuitable for an embedded real-time context. but this project's approach lightweight, embedded, real-time is genuinely underserved in the literature, so the positioning makes sense. The evaluation plan also looks well-structured, covering idle, normal, heavy, and I/O-bound workload scenarios. The main open question at this stage is whether the SPI arbitration logic holds up under actual stress testing, since that's the most unpredictable part of the system.

Group 5

Metin Irkıcatal, Anıl Arda

Project Title: *Edge-AI Based Smart Access Control System*



Group 6

Oğulcan Uğuroğlu, Barış Büyükyılmaz

Project Title: *Contention-Aware Duplication Scheduling for NoC Systems*

Group 6's project is about contention-aware duplication scheduling for Network-on-Chip based multicore systems. The main idea is to improve task scheduling by deciding not only whether duplicating a task reduces execution time, but also whether that duplication will increase network congestion. I found this motivation technically strong because NoC-based multicore systems are used in performance-sensitive embedded domains such as autonomous vehicles, medical imaging, industrial automation, and telecommunications. In these systems, poor scheduling can increase communication latency, energy consumption, and even cause missed real-time deadlines.

The strongest part of the proposal is that it clearly defines a comparison between three scheduling strategies: baseline HEFT without duplication, classical duplication-based scheduling, and the proposed contention-aware heuristic. Their plan to use synthetic DAGs from TGFF or daggen and evaluate network behavior with BookSim 2.0 also makes the project measurable and realistic. The proposed penalty term based on link utilization, hop count, and buffer pressure is a good idea because it directly connects scheduling decisions with actual NoC congestion. A few things were less clear. Since the project is an optimization-track project, the final success will depend heavily on how the contention penalty is formulated and tuned. It would also be useful to explain whether the heuristic works well only for synthetic DAGs or if it can generalize to more realistic embedded workloads. Also, energy overhead is mentioned, but it is not fully clear how energy will be measured or estimated.

Overall, this is a solid and technically focused project. It has a clear problem, strong motivation, relevant related work, and measurable metrics such as makespan, link utilization balance, scheduling fairness, and duplication overhead. The project would be stronger if the evaluation includes different NoC sizes and CCR values clearly, and if the proposed heuristic is explained with a simple example.

Group 7

Ahmet Yiğitalp Demirci

Project Title: *Real-Time Energy-Constrained UAV Optimization*

Group 8

Sami Enes Yılmaz, Kadir Emre Avcı

Project Title: *SmartAquaria System*

Group 9

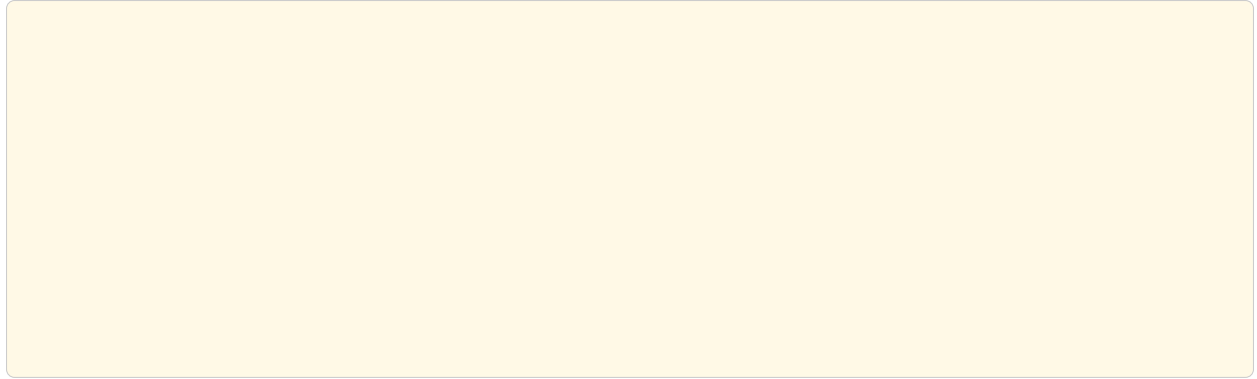
Mehmethan Ayrım

Project Title: *Energy-Aware Mixed-Criticality Scheduling*

Group 10

İrem Gül Subaşı

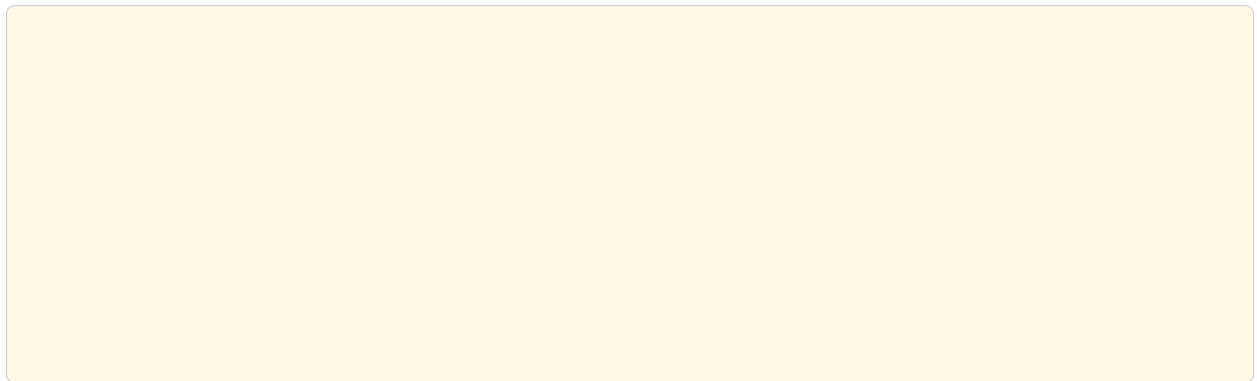
Project Title: *Fault Injection and Resilience Profiling of Lightweight Transformers*



Group 11

Süleyman Ege Karakaya

Project Title: *Adaptive Data-Movement Strategy for Crypto Accelerators*



Group 12

Gökçer Erol

Project Title: *Acoustic Localization System for Search and Rescue*

Group 12's project is about acoustic localization for search and rescue after disasters. The goal is to detect survivor sounds such as knocking or whistling using low-cost embedded sensor nodes, then estimate the survivor's 2D position with Time Difference of Arrival, or TDoA. I found the motivation strong because in collapsed buildings, visual access, Wi-Fi, and normal communication can be limited, while acoustic signals may still travel through debris. The system uses ESP32-S3 sensor nodes with INMP441 microphones, GNSS PPS timing, LoRa communication, and a Jetson Nano gateway, which makes the design realistic for field deployment.

The strongest part of the project is that it considers real embedded constraints. The nodes do not stream raw audio; instead, they detect sound, use TinyML to reject noise, timestamp the event, and transmit a compact LoRa packet with metadata and CRC. This is a smart design choice because LoRa bandwidth is limited. The proposal also gives clear targets such as under 10 μ s clock drift, 48-byte payloads, 24-hour battery life, and under 2 seconds end-to-end latency. The central hub localization pipeline is also well planned.

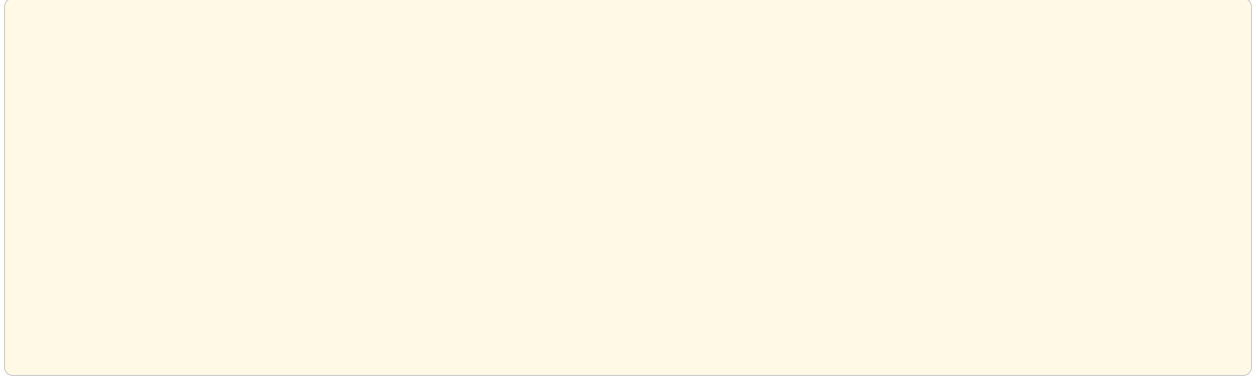
A few points were less clear. The project assumes constant sound speed and relatively ideal acoustic propagation, but real rubble can cause reflection, attenuation, and multipath, which may reduce localization accuracy. Also, the system relies on GNSS PPS synchronization, but GNSS may not always work well near or inside collapsed structures.

Overall, this is a strong and realistic embedded systems project. It combines sensing, low-power communication, timing synchronization, TinyML filtering, and localization in a meaningful way. The project would be stronger if the team shows how robust the system is under noisy rubble-like conditions and weak GNSS availability.

Group 13

Emre Doğan, Yavuz Selim Kızıltaş

Project Title: *Optimization of Edge AI Pipeline on TDA4VM*



Group 14

Mehmet Yusuf Bilge, Sezil Ağırbaş

Project Title: *Closed-Loop Vestibular Stimulation System*

Group 14's project is about a wearable system that helps people with balance disorders by stimulating the vestibular system — the part of the inner ear responsible for balance — using small electrical currents applied behind the ear. The idea is that when someone starts to lose balance, the system detects it through an IMU sensor, runs that through a control algorithm, and delivers corrective stimulation in real time to help the person stabilize. The closed-loop nature of this is what makes it interesting and also what makes it genuinely difficult.

The system architecture is clean and well thought out. An IMU reads orientation data, which gets processed through a sensor fusion filter, fed into a PID controller running on an MCU under FreeRTOS, and the output drives a constant current stimulation circuit. The separation between sensing, processing, and actuation is clear, and the inclusion of a safety monitor as a dedicated component rather than an afterthought shows good design judgment for a system that is ultimately touching a human body.

The safety aspect is the most critical part of this project, and from what was presented it was taken seriously. Constant current control rather than voltage control is the right choice for body-interfacing systems since tissue resistance varies and a fixed voltage could deliver unpredictable current. Hardware-level current limiting on top of software watchdogs is the kind of layered protection you want in a safety-critical system.

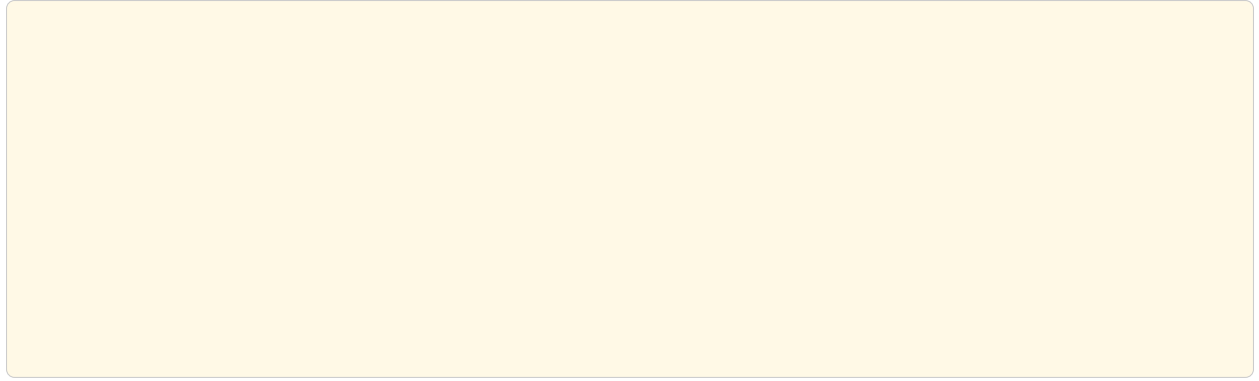
A few things I would have liked to see more clearly addressed. The maximum allowable latency between sensing and stimulation was not explicitly quantified for a balance assistance system this matters a lot. Similarly, the FreeRTOS task structure and how sampling, control, and safety monitoring are scheduled relative to each other was not fully detailed. I'd also be curious about how the system behaves during sensor noise spikes or when the IMU gives ambiguous readings.

Overall it's a well-motivated project with a clear real-world application and a solid embedded system foundation. The main open questions are around formalizing the timing model and demonstrating that the safety constraints actually hold under stress conditions, which seem to be planned for the later phases.

Group 15

Ayberk Aygün, Berkay Taşkın

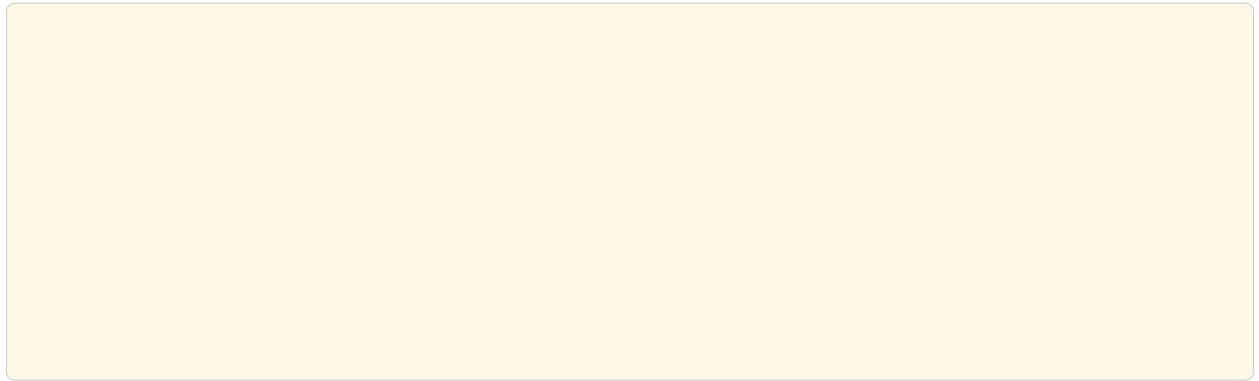
Project Title: *Quadrotor Simulation on NVIDIA Jetson*



Group 16

Yasin Arslan, Süleyman Sertaç Üst

Project Title: *Real-Time Audio Pipeline Analysis on Heterogeneous SoCs*



Group 17

Oğuz Kağan Yağlıoğlu, Emre Eşef Kılıçdoğan

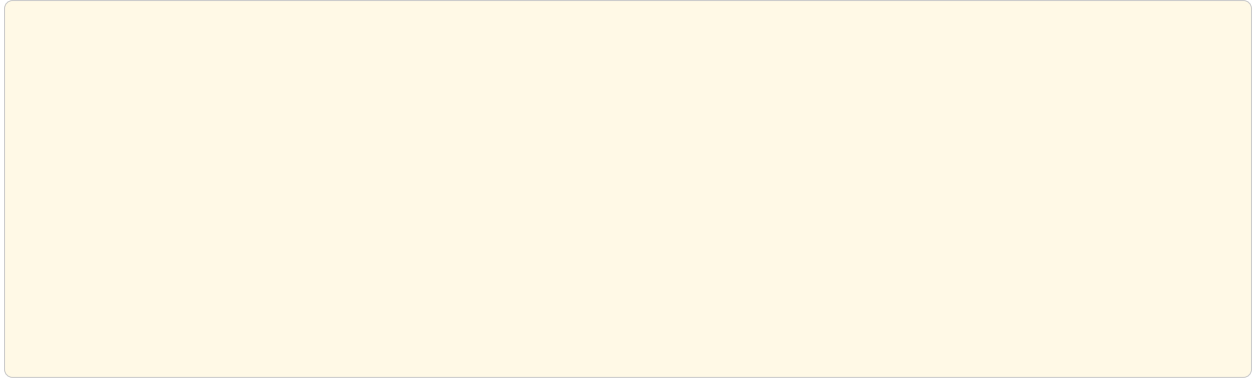
Project Title: *Predictive Maintenance of BLDC Motors via Multi-Sensor Fusion*

Group 17's project is about detecting faults in BLDC motors before they actually break down. The idea is to attach a small sensor device to a motor, collect data from it, and use machine learning to figure out if something is going wrong and if so, what kind of fault it is. The project covers both hardware design and the ML side, and I found it well thought-out overall. The device includes an accelerometer, a temperature sensor, and an ultrasonic microphone which, as far as I understood, helps catch certain fault types that a regular microphone would miss. It communicates over LoRa, a long-range low-power wireless protocol suitable for industrial environments where WiFi might not be available. The enclosure is 3D printed with carbon fiber-reinforced material and is IP67 rated, making it dust and water resistant. The ML part was the most impressive to me. They showed how they improved step by step, from 66% to 92% accuracy. The biggest jump came from merging three fault types that were too similar for the model to tell apart which I thought was a smart and honest decision rather than forcing the model to do something it couldn't reliably do. The other key insight was that the same fault looks different to the model depending on the motor's load and speed. Once they added that information as an input, cross-load accuracy jumped from around 44% to 92%. A few things were less clear after the presentation. The device compresses data into a very small packet for LoRa transmission, but the model uses 30+ features I wasn't entirely sure how those two things reconcile. Also, the model was trained on a dataset from a different motor than the one they are building for, so how well it transfers to their own hardware remains an open question. They acknowledged this and listed it as a future plan, which seems reasonable. Overall, it is a genuinely useful and realistic project. The hardware looks well-designed, the ML pipeline shows real iteration and problem-solving, and the goal of catching motor faults before they cause downtime is clearly motivated.

Group 18

Hüseyin Eren Doğan

Project Title: *EdgeRAG: Offline RAG on Embedded Systems*



Group 19

Ali Bölücü

Project Title: *Spectral-Seeded NSGA-II for Many-Core Systems*

Group 19's project tackles the problem of deciding which task runs on which core in a many-core chip. This sounds simple but is actually an astronomically large combinatorial problem, the number of possible assignments grows factorially, making it impossible to search exhaustively. Bad mappings lead to tasks that communicate heavily being placed far apart on the chip, increasing latency and creating thermal hotspots where certain areas of the chip overheat while others sit idle. The issue is that NSGA-II starts with a random population and wastes many generations just getting to a reasonable region of the solution space. This project's core idea is to give it a head start: instead of initializing everything randomly, 20 percent of the initial population is seeded with solutions pre-computed using spectral graph partitioning. The spectral method analyzes the communication graph of the tasks, finds natural clusters of heavily-communicating tasks, and groups them together. These clusters then become the "elite seeds" that guide the evolutionary search from a much better starting point. The pipeline is well thought out. First all application graphs are merged into a single unified affinity matrix. Then spectral decomposition via the Fiedler vector finds natural task clusters. The silhouette coefficient is used to automatically determine the optimal number of clusters rather than fixing it manually. Finally these clustered seeds are mixed with 80 percent random individuals to preserve diversity before NSGA-II runs. The 80/20 split is a reasonable balance too many seeds and you lose diversity, too few and the benefit disappears. Compared to prior works in the related work table, this project is one of the few that combines spectral clustering, dynamic K selection via silhouette, thermal awareness, and multi-application support all together using NSGA-II, which is a meaningful combination. The closest prior work is about with thermal awareness and multi-app support but without the spectral seeding which is exactly the gap this project addresses. A few things I wasn't entirely sure about after the presentation. The evaluation compares against random NSGA-II and pure spectral mapping, but I'd be curious whether the improvement in convergence speed is consistent across different application mixes, or if certain task graph structures benefit more than others. Also, since this is an offline mapping tool run before deployment, the computational cost of the spectral pre-processing itself wasn't discussed but i think for an offline tool this is probably acceptable. The results are still expected, so the actual improvement figures remain to be seen.

Group 20

İlter Doğaç Sönmez

Project Title: *Autonomous Obstacle Avoidance Vehicle using LiDAR*

Group 20's project is about building and emulating an autonomous obstacle avoidance vehicle using LIDAR and an embedded system. The project is mainly based on the paper "Autonomous Obstacle Avoidance Vehicle Using LIDAR and an Embedded System," which proposes a Raspberry Pi based miniature vehicle that can navigate indoors using a single LIDAR sensor instead of computer vision, making it more suitable for low-light environments. The presentation explains this idea clearly and connects it to an embedded system design problem: testing the behavior of the vehicle before deploying it on real hardware. The hardware side includes a Raspberry Pi 3, an RPLidar A2 sensor, a motor controller, a steering motor, a main motor, and battery/power components, which makes the system realistic rather than purely theoretical. The most interesting part for me was the emulation idea. Instead of only building the vehicle physically, they aim to create a testing environment where the embedded hardware, GPIO pins, peripherals, kernel/OS, and application code can be modeled in a host environment. This fits well with the motivation of catching problems earlier, before they become harder and more expensive to fix.

The related work section is also relevant because it compares the project with newer autonomous vehicle and robotic perception systems, including ML-based electric buggy control, visual-LiDAR fusion with SLAM, and Ackermann steering robot localization using RGB camera and 2D LIDAR. These references show that the topic is still active and that LIDAR-based navigation is commonly combined with embedded platforms and real-time control. However, one thing that was less clear after the presentation is how detailed the emulation will actually be. For example, the slides show a Renode-style architecture, but it is not fully explained how the LIDAR readings, motor behavior, and physical environment will be realistically simulated. Another unclear point is the difference between reproducing the original paper's vehicle logic and extending it with a new emulation framework. The motivation slide also focuses strongly on general defect cost in production, but it could be connected more directly to autonomous vehicle testing, sensor faults, or embedded control failures.

Overall, I think this is a useful and realistic embedded systems project. The hardware architecture is understandable, the main paper is appropriate for a Raspberry Pi and LIDAR-based autonomous vehicle, and the emulation direction makes the project more valuable than simply assembling a robot. The project would be stronger if the team clearly defines what exactly will be emulated, what will remain physical, and how they will evaluate success, such as obstacle avoidance accuracy, response time, collision rate, or comparison between emulated and real hardware behavior. Based on the presentation, the project has a solid foundation, but the implementation details of the emulation and validation methodology need to be explained more clearly.